

Jin Yang, Ph.D Candidate.

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🐙 Github



Education

Sep. 2019 – Aug. 2026 (expected)

Combined M.S - Ph.D in Computer Science

Institution: Ulsan National Institute of Science and Technology (UNIST), Korea.

Advisor: Prof. Young-ri Choi.

Mar. 2015 – Aug. 2019

B.S in Computer Science and Engineering (major) and Mathematics (minor)

Institution: Ulsan National Institute of Science and Technology (UNIST), Korea.

Research interests

Storage Systems

Indexing Structure

Key-Value Store

Data Management in Storage Engines

Research Publications

Conference Proceedings

- 1 **Jin Yang**, Heejin Yoon, Gyeongchan Yun, Sam H. Noh, and Young-ri Choi, “DyTIS: A Dynamic Dataset Targeted Index Structure Simultaneously Efficient for Search, Insert, and Scan,” in *Proceedings of the 18th European Conference on Computer Systems*, 2023, pp. 800–816. [DOI: 10.1145/3552326.3587448](#).
- 2 Heejin Yoon, **Jin Yang**, Juyoung Bang, Sam H. Noh, and Young-ri Choi, “Advocating for Key-Value Stores with Workload Pattern Aware Dynamic Compaction,” in *Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '24)*, 2024. [DOI: 10.1145/3655038.3665955](#).

Journal Articles

- 1 **Jin Yang**, Heejin Yoon, Gyeongchan Yun, Sam H. Noh, and Young-ri Choi, “A Dynamic Characteristic Aware Index Structure Optimized for Real-world Datasets,” *ACM Transactions on Storage*, vol. 21, no. 2, pp. 1–32, 2025. [DOI: 10.1145/3707642](#).

Patents

- 1 Young-ri Choi, **Jin Yang**, Heejin Yoon, Gyeongchan Yun, and Sam H. Noh, *Method and Apparatus for Performing Dynamic Hashing to Store Key Values in an Ordered State*, KR Patent 10-2683404 (granted Jul. 2024); US Patent App. Pub. US 2024/0048358 A1 (Feb. 2024), 2024.
- 2 Young-ri Choi, Songyi Lee, **Jin Yang**, and Sam H. Noh, *Method and Apparatus for Converting Data Key Value*, KR Patent 10-2388458 (granted Apr. 2022), 2022.

GitHub Projects

- DyTIS** An open-source repository for "DyTIS: A Dynamic Dataset Targeted Index Structure Simultaneously Efficient for Search, Insert, and Scan" accepted at EuroSys '23.
(<https://github.com/unist-ssl/DyTIS>)
- HASS** A repository for Hotness-Aware Sharding Split for MongoDB.
(<https://github.com/heejin5178/HASS>)
- RH-Skiplist** A repository for indexing structure with skiplist and hash for hybrid memory system with Intel Optane persistent memory and DRAM.
(<https://github.com/sheepjin11/RH-Skiplist>)

Awards and honors

- 2023 **EuroSys 2023 Grant**, EuroSys Conference
- 2018 **UNIST Hackathon Alkkago Award**, sponsored by Naver D2SF (500,000 KRW)
- 2016 **UNIST HeXATHON Finalist (Top 12 teams)**, sponsored by Naver D2 and UNIST ECE

Skills

- Programming C++, C, Python.
- Languages English and Korean.

Work Experiences

- Mar. 2022 - Aug. 2022 Teaching assistant, Introduction to Human Computer Interaction, undergraduate course at UNIST.
- Sep. 2021 - Feb. 2022 Teaching assistant, Data Structures, undergraduate course at UNIST.
- Mar. 2021 - Aug. 2021 Teaching assistant, Advanced Programming, undergraduate course at UNIST.
- Sep. 2020 - Feb. 2021 Teaching assistant, Operating Systems, undergraduate course at UNIST.
- Mar. 2020 - Aug. 2020 Teaching assistant, Computer Networks, undergraduate course at UNIST.
- Sep. 2019 - Feb. 2020 Teaching assistant, Introduction to AI Programming 2, undergraduate course at UNIST.
- Mar. 2019 - Dec. 2019 7th Organizer, Like Lion - a nonprofit organization focused on programming and web development, UNIST
- Mar. 2018 - Dec. 2018 6th Organizer, Like Lion - a nonprofit organization focused on programming and web development, UNIST
- Dec. 2017 - Feb. 2018 Intern Developer at DoDotDo.
- Mar. 2017 - Dec. 2017 5th Member at Like Lion - a nonprofit organization focused on programming and web development, UNIST
- Sep. 2016 - Feb. 2017 Teaching assistant, Linear algebra, undergraduate course at UNIST.